

Yuchen Hou

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Summary

Master student at the National University of Singapore, focusing on Embodied AI and autonomous robotic decision-making.

Education

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| <p>M.Sc. National University of Singapore, Electrical Engineering</p> <ul style="list-style-type: none"> Core member of the NUS Advanced Control Lab, focusing on Embodied AI and autonomous robotic decision-making. Research on perception and planning for aerial robots based on Vision-Language Navigation (VLN), and end-to-end Reinforcement Learning in complex 3D environments. | <p>Singapore
Aug 2025 – present</p> |
| <p>B.Eng. Nanjing University of Aeronautics and Astronautics, Electronic and Communication Engineering</p> <ul style="list-style-type: none"> GPA: 88/100 Key Courses: Advanced Mathematics (96), Probability Theory (97), Random Signal Analysis (97), Microwave Technology and Antenna (97), Fundamentals of Communication (95), Complex Function (99). | <p>Nanjing, China
Sept 2021 – July 2025</p> |

Experience

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| <p>Honor Device Co., Ltd., VLA Algorithm Engineer (Intern)</p> <ul style="list-style-type: none"> Working on Honor's self-developed Omega humanoid robot: teleoperation data collection, replay, training, and on-robot deployment. Conducting online RL algorithm research for Vision-Language-Action (VLA) models on custom manipulation tasks. | <p>Shanghai, China
June 2026 – present
3 months</p> |
| <p>Huawei Norbert Wiener Research Center, Research Intern</p> <ul style="list-style-type: none"> Developed backend deep learning models for short-range radar signal processing. Conducted research on downstream tasks including human pose estimation and indoor mapping/SLAM from radar data. | <p>Singapore
Mar 2026 – May 2026
3 months</p> |
| <p>Movel.AI, Robotics Software Intern</p> <ul style="list-style-type: none"> Validated autonomous navigation and control systems for forklift robots in industrial environments. Conducted on-site factory deployment and testing in real warehouse settings. Integrated robotics software with fleet management systems. Tech Stack: ROS2, Navigation Stack, Sensor Integration. | <p>Singapore
Dec 2025 – Jan 2026
2 months</p> |

Awards

- Second-class Scholarship:** Nanjing University of Aeronautics and Astronautics (2024)
- Third-class Scholarship:** Nanjing University of Aeronautics and Astronautics (2021 – 2023)

Skills

Programming & Systems: Python (ROS 2, PyTorch, OpenCV), Linux (Ubuntu), Shell scripting

AI Frameworks & Simulation: PyTorch (DRL policies), Habitat, AI2-THOR (Complex 3D environments)

Robotics Development: ROS2 (rclpy, Nav2), Sensor Fusion, Lidar/RGB-D Integration, SLAM (Cartographer)

Research Capabilities: Sim-to-Real Transfer, Embodied AI Architectures, RL Algorithms, Academic Paper Writing, Tracking
CVPR/IROS/ICRA

Languages

English: Proficient — IELTS 7.0 (R 8.5, W 7.5, L 6.5, S 6.0)

Mandarin: Native speaker